

Shape-Preserving Approximation *

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Here, “shape” means the curvature and monotonicity properties of a function. We can use piecewise-linear interpolation to preserve positivity, monotonicity, and concavity, but it is not differentiable. So we want to produce a smooth function with shape-preserving quadratic spline of Schumaker [3].

The basic Hermite problem on the single interval $[t_1, t_2]$ takes the data z_1, z_2, s_1, s_2 and constructs a piecewise-quadratic function $s \in C^1[t_1, t_2]$ such that

$$s(t_i) = z_i, \quad s'(t_i) = s_i, \quad i = 1, 2. \quad (1)$$

LEMMA 1. If $\frac{s_1+s_2}{2} = \frac{z_2-z_1}{t_2-t_1}$, then the quadratic function

$$s(t) = z_1 + s_1(t - t_1) + \frac{(s_2 - s_1)(t - t_1)^2}{2(t_2 - t_1)} \quad (2)$$

satisfies (1).

You may want to check the shape-preserving properties of this construction.

But lemma 1 does not apply in general. Another strategy is to add a knot to the interval $[a, b]$ and then construct a spline with the desired properties using the following lemma.

LEMMA 2. For every $\xi \in (t_1, t_2)$, there is a unique quadratic spline solving (1) with a knot at ξ , which is

$$s(t) = \begin{cases} A_1 + B_1(t - t_1) + C_1(t - t_1)^2, & t \in [t_1, \xi] \\ A_2 + B_2(t - \xi) + C_2(t - \xi)^2, & t \in [\xi, t_2]. \end{cases} \quad (3)$$

where

$$\begin{aligned} A_1 &= z_1, \quad B_1 = s_1, \quad C_1 = \frac{\bar{s} - s_1}{2\alpha}, \\ A_2 &= A_1 + \alpha B_1 + \alpha^2 C_1, \quad B_2 = \bar{s}, \quad C_2 = \frac{s_2 - \bar{s}}{2\beta}, \\ \bar{s} &= s'(\xi) = \frac{2(z_2 - z_1) - (\alpha s_1 + \beta s_2)}{t_2 - t_1}, \\ \alpha &= \xi - t_1, \quad \beta = t_2 - \xi. \end{aligned}$$

*For more discussions, please refer to the textbooks and the lecture notes of Professor Zhang. Comments are welcome.

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Note that ξ is a free node in that we do not know $s(\xi)$, which is allowed to be free chosen as well.

LEMMA 3 (MONOTONICITY) Suppose $s_1 \cdot s_2 \geq 0$. Then the spline (3) is monotone on $[t_1, t_2]$ if and only if $s_1 \cdot \bar{s} \geq 0$. This condition can also be written as

$$\begin{aligned} 2(z_2 - z_1) &\geq (\xi - t_1)s_1 + (t_2 - \xi)s_2 & \text{if } s_1, s_2 \geq 0 \\ 2(z_2 - z_1) &\leq (\xi - t_1)s_1 + (t_2 - \xi)s_2 & \text{if } s_1, s_2 \leq 0 \end{aligned}$$

For curvature, notice the following facts: i. if $s_1 < s_2$, the spline s in (3) is convex on $[t_1, t_2]$ if and only if $s_1 \leq \bar{s} \leq s_2$; ii. if $s_1 > s_2$, then s is concave on $[t_1, t_2]$ if and only if $s_2 \leq \bar{s} \leq s_1$. So, we need to find the knot leading to convex or concave splines, which is indicated by the following lemma.

LEMMA 4 (CURVATURE) Let $\delta = \frac{z_2 - z_1}{t_2 - t_1}$. Then $(s_2 - \delta)(s_1 - \delta) \geq 0$ implies that s must have an inflection point in the interval $[t_1, t_2]$, which means that we have neither a concave nor convex interpolant. Suppose now that $(s_2 - \delta)(s_1 - \delta) < 0$. Then:

- If $|s_2 - \delta| < |s_1 - \delta|$, for all ξ satisfying

$$t_1 < \xi \leq \bar{\xi} \quad \text{with} \quad \bar{\xi} = t_1 + \frac{2(t_2 - t_1)(s_2 - \delta)}{s_2 - s_1} \quad (4)$$

the interpolating spline s in (3) satisfies s is convex (concave) on $[t_1, t_2]$ if $s_1 < s_2$ ($s_1 > s_2$). Moreover, if $s_1 s_2 \geq 0$ then s is also monotone.

- If $|s_2 - \delta| > |s_1 - \delta|$, for all ξ satisfying

$$\underline{\xi} \leq \xi < t_2 \quad \text{with} \quad \underline{\xi} = t_2 + \frac{2(t_2 - t_1)(s_1 - \delta)}{s_2 - s_1} \quad (5)$$

the interpolating spline s in (3) satisfies s is convex (concave) on $[t_1, t_2]$ if $s_1 < s_2$ ($s_1 > s_2$). Similarly, if $s_1 s_2 \geq 0$ then s is also monotone.

Using the above results, we have the algorithm as follows.

1. Check whether lemma 1 holds. If so, use the result of lemma 1 to get $s(t)$, and STOP.
2. Compute $\delta = \frac{z_2 - z_1}{t_2 - t_1}$.
3. Check the value of $(s_1 - \delta)(s_2 - \delta)$.
 - If $(s_1 - \delta)(s_2 - \delta) \geq 0$, set $\xi = 1/2(t_1 + t_2)$ and STOP.
 - If $(s_1 - \delta)(s_2 - \delta) < 0$, check whether $|s_2 - \delta| < |s_1 - \delta|$.
 - If $|s_2 - \delta| < |s_1 - \delta|$, compute $\bar{\xi}$ using (4), set $\xi = 1/2(t_1 + \bar{\xi})$ and STOP.
 - If $|s_2 - \delta| \geq |s_1 - \delta|$, compute $\underline{\xi}$ using (5), set $\xi = 1/2(t_2 + \underline{\xi})$ and STOP.

For a general interpolation problem, if we have Hermite data, $\{(z_i, s_i, t_i) | i = 1, \dots, n\}$, we apply the shape-preserving interpolant algorithm to each interval to find $\xi_i \in [t_i, t_{i+1}]$. If we have Lagrange data, $\{(z_i, t_i) | i = 1, \dots, n\}$, we must first estimate the slopes to get Hermite problem. Schumaker suggests the following methods to estimate slopes s_1 through s_n :

- If $i = 1$, $s_1 = -\frac{3\delta_1 - s_2}{2}$;
- For $i = 2, \dots, n-1$,

$$s_i = \begin{cases} \frac{L_{i-1}\delta_{i-1} + L_i\delta_i}{L_{i-1} + L_i} & \text{if } \delta_{i-1}\delta_i > 0, \\ 0, & \text{otherwise.} \end{cases}$$

- If $i = n$, $s_n = \frac{3\delta_{n-1} - s_{n-1}}{2}$.

where,

$$L_i = [(t_{i+1} - t_i)^2 + (z_{i+1} - z_i)^2]^{\frac{1}{2}}, \quad i = 1, \dots, n-1$$

$$\delta_i = \frac{z_{i+1} - z_i}{t_{i+1} - t_i}, \quad i = 1, \dots, n-1.$$

References

- [1] Judd, Kenneth L. 1998. *Numerical Methods in Economics*, The MIT Press.
- [2] Judd, Kenneth L. and Solnick, Andrew. 1994. "Numerical Dynamic Programming with Shape-Preserving Splines". Mimeo.
- [3] Schumaker, Larry L. 1983. "On Shape-Preserving Quadratic Spline Interpolation". *SIAM Journal of Numerical Analysis* 20: 854-64.